

Task 1 — Document 04: Target Users & Industrial Personas

Problem Statement ID: PS 26117 / SIH26117 **Title:** Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work **Organization:** Mangalore Refinery and Petrochemicals Limited (MRPL) **Category:** Software | **Theme:** Smart Automation

■ Overview of Target User Groups

The **Sovereign AI Workbench** is tailored for multidisciplinary teams operating within confidential industrial refining facilities. Below are the primary user personas, their operational pain points, and specific workbench use cases.

Persona Group	Primary Focus & Tasks	Primary Workbench Solution
■ ■ 1. Plant Maintenance & Equipment Engineers	Pump inspection verification, compressor SOP step retrieval, maintenance logs.	Grounded RAG search over equipment SOP manuals & OCR scanned inspection sheets.
■ ■ 2. Health, Safety & Environment (HSE) Officers	Emergency shutdown procedures, hazardous chemical handling, compliance audits.	Grounded safety manual retrieval with strict anti-hallucination fallback.
■ 3. Industrial IT & Automation Engineers	Writing automation scripts, PLC data parsing, API development, local debugging.	Local code synthesis via qwen2.5-coder:7b without cloud code exposure.
■ 4. Plant Management & Executive Leadership	Decision support, executive shift report summaries, milestone tracking.	Instant AI summarization of daily shift reports and plant progress logs.

■ ■ Detailed Persona Profiles

Persona 1: Plant Maintenance & Equipment Engineers

- **Role:** Field Engineers responsible for maintaining mechanical equipment (pumps, turbines, heat exchangers, valves) across MRPL refinery units.
- **Pain Point:** Finding specific torque values, vibration limits, or inspection checklists buried in 500-page vendor manuals or handwritten paper logs.
- **Workbench Solution:**
 - Uploads equipment manual PDFs and photos of handwritten inspection sheets.
 - Queries: "What is the acceptable pump vibration threshold during normal operation according to the SOP?"
 - Receives grounded answers with direct citations (**test_sop.txt**).

Persona 2: Health, Safety & Environment (HSE) Officers

- **Role:** Officers ensuring strict adherence to environmental regulations, personal protective equipment (PPE) guidelines, and emergency shutdown protocols.
- **Pain Point:** Cross-referencing multiple safety regulations during live plant safety audits or incident investigations.
- **Workbench Solution:**
 - Indexes all plant safety manuals into the local Knowledge Base.
 - Queries: "What are the mandated safety protocols for hydrogen sulfide (H2S) leak containment?"
 - Relies on anti-hallucination guardrails: if an H2S manual hasn't been uploaded, system responds with missing data warning rather than inventing safety steps.

Persona 3: Industrial IT & Automation Engineers

- **Role:** Control system engineers and software developers writing internal automation scripts, PLC data parsers, and monitoring tools.
- **Pain Point:** Needing AI coding assistance for Python/FastAPI automation without leaking internal code snippets to commercial cloud LLMs.
- **Workbench Solution:**
 - Uses the workbench for code generation and debugging.
 - Automatic model router routes coding prompts to **qwen2.5-coder:7b**.
 - Generates clean Python code, regex parsers, or SQL queries locally.

Persona 4: Plant Management & Executive Leadership

- **Role:** Refinery Operations Managers, Unit Heads, and General Managers.
- **Pain Point:** Requiring immediate operational insights from daily plant shift logs and high-level reports without waiting for manual summary compilation.
- **Workbench Solution:**
 - Uploads daily shift reports and milestone progress documents.
 - Asks high-level summary questions: *"Summarize key maintenance activities completed during Shift A in Unit 2."*
 - Fast, grounded decision support with zero data privacy risks.

■ Business Impact & Benefit Matrix

Persona	Primary Metric Improved	Before Workbench	With Sovereign Workbench
Maintenance Engineer	SOP Retrieval Time	45-60 minutes manual search	< 5 seconds grounded answer
HSE Officer	Audit Accuracy & Safety Compliance	Manual cross-referencing paper binders	100% Grounded RAG with source citations
Automation Engineer	Code Generation Speed	2-3 hours writing scripts from scratch	Instant local code synthesis via Qwen Coder
Plant Manager	Decision Support Velocity	Hours waiting for compiled shift reports	Instant AI summarization of local shift logs
Cybersecurity Team	Data Leakage Prevention	Risk of sensitive data sent to Cloud APIs	100% Air-Gapped, 0 outbound calls